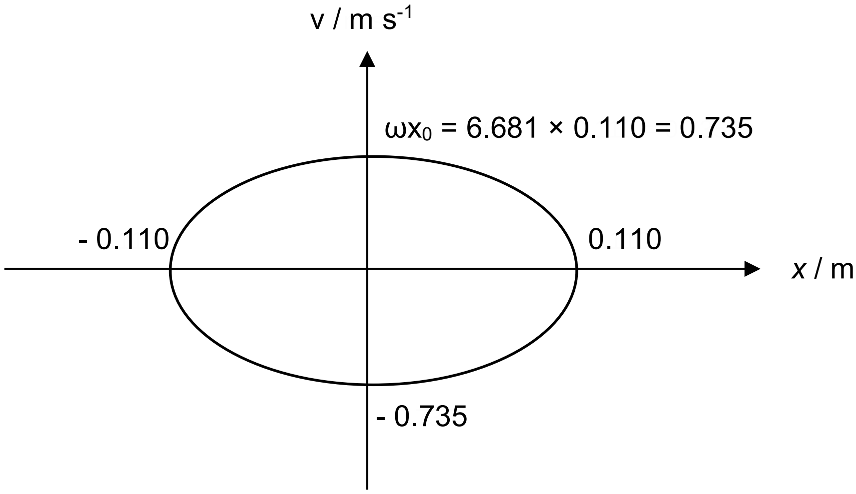
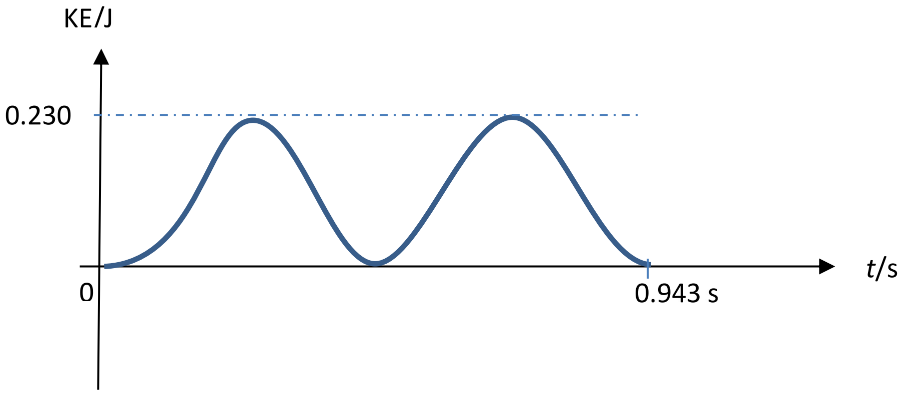
2	(a)	At equilibrium, the net force is zero hence weight = spring force $mg = kx$ $0.850(9.81) = k(0.220)$ [1] $k = \dfrac{0.850(9.81)}{(0.2200)} = 37.9$ N m^{-1}		
	(b)	(i)	F_{net} = spring force – weight $= ke - mg$ $= 37.9(0.110 + 0.220) - 0.850(9.81)$ $= 4.17$ N $F_{net} = ma$ $a = \dfrac{Fnet}{m} = \dfrac{37.9(0.330) - 0.850(9.81)}{0.850}$ [1] $a = 4.91$ m s^{-2} [1]	
		(ii)	$a_0 = \omega^2 x_0$ $4.91 = \omega^2 (0.110)$ $\omega^2 = \dfrac{4.91}{0.110}$ $\omega = 2\pi f$ $f = \dfrac{\omega}{2\pi} = \dfrac{\sqrt{\frac{4.91}{0.110}}}{2\pi}$ [1] $f = 1.06$ Hz [1]	

(iii)



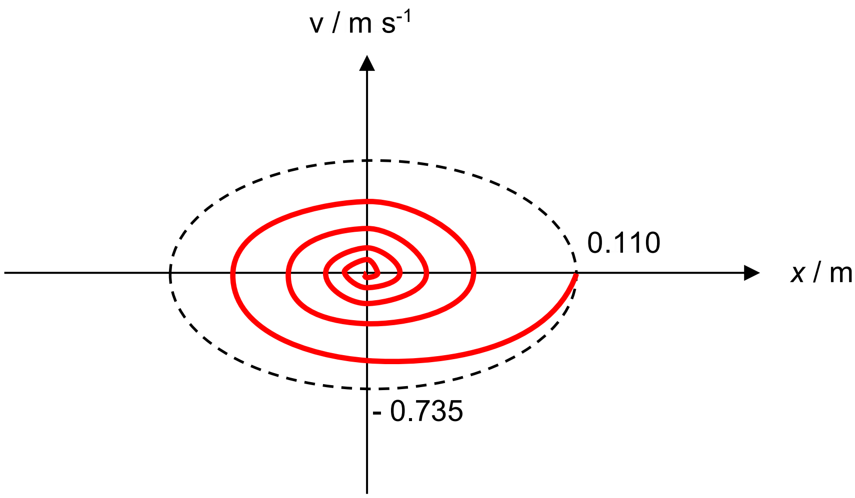
[1] shape must be an ellipse
[1] correct values of x-intercepts and y-intercepts of ellipse.

(iv)



[1] Sine-squared wave in one period
[1] Labels: $KE_{max} = \frac{1}{2} m\omega^2 x_0^2 = 0.5(0.850)(44.636)(0.110)^2 = 0.230$ J
and period = 1/f = 1/1.06 = 0.941 s or 0.943 s.

(v)



[1] Spiral clockwise inwards starting from at x_0 = + 0.110 m.
[1] Exponentially decreasing amplitude of oscillation (spacing between x-intercepts become smaller), at least 3 complete spiral rounds